

A Comparative Study of Thermodynamic Efficiency and Projected CO2 Intensity in USC and A-USC Power Plants in Croatia

Scholar Name: SK JEZZBAH ISLAM

Scholar Institute Name: Presidency International School

Date of Submission: 1st May 2026

Abstract

on Supercritical and Advanced-Ultra Supercritical power plants and how transitioning to the more advanced ones helps mitigate the drawbacks. This research presents a comparative-quantitative study of thermodynamic efficiency and intensity of CO₂ emitted in Supercritical (SC) and Advanced Ultra-Supercritical (A-USC) power plants regulated in Croatia. By using secondary data sources, this study aims to determine the quantifiable gain in the net efficiency, which results in the reduction of the CO₂ emission, as efficiency and CO₂ emission are inversely proportional.

By using purposive sampling, the research analyses how different steam parameters influence the performance of a plant. A hypothesis suggests that Advanced Ultra-Supercritical power plants exhibit a higher percentage of net efficiency, approximately 47.6%, which represents an improvement of 8% than other older plants, while significantly lowering the intensity of CO₂ emitted into the atmosphere, to 700 kg CO₂/MWh. Additionally, the study quantifies the impact of Carbon Capture and Storage (CCS). Despite reducing emissions by nearly 90%, it reduces the net efficiency by 10.8%.

The final result indicates that the higher baseline efficiency of A-USC technology is crucial to mitigate the energy penalty associated with decarbonisation, confirming the hypothesis that A-USC plants are the most viable pathway for balancing higher power output with stringent environmental targets.

Introduction

The global energy sector faces a challenge in meeting rising electricity demand. for electricity while drastically reducing greenhouse gas emissions into the atmosphere. Coal remains a significant part of this procedure. This study focuses